AP PRECALCULUS

Unit 1 • Lesson 1.1 practice

Lesson 1.1 • Change in Tandem

4 multiple-choice questions and 1 free-response question

Question 1

The table gives values of f. On which interval does the table show f decreasing?

x	-2	-1	0	1	2
$f(x)$	7	2	-1	-2	-1

(A) $[-2, 1]$

(B) $[-1, 2]$

(C) $[0, 2]$

(D) $[-2, 2]$

Question 2

What are the zeros of $g(x) = (x - 4)(x + 1)$?

(A) -4 and 1

(B) 4 only

(C) -1 and 4

(D) 0 and 4

Question 3

A function is increasing and its rate of change is decreasing. Which describes its graph?

(A) Increasing and concave up

(B) Increasing and concave down

(C) Decreasing and concave up

(D) Decreasing and concave down

Question 4

A car's distance d (meters) from a stop sign is recorded: $d(0) = 0$, $d(1) = 10$, $d(2) = 40$, $d(3) = 90$. Which statement fits the data?

(A) d is decreasing and concave up.

(B) d is increasing and concave down.

(C) d is linear.

(D) d is increasing and concave up.

Question 5 (Free response)

The height h, in meters, of a ball is recorded every half second.

t (s)	0	0.5	1	1.5	2
$h(t)$ (m)	2	9	12	11	6

(a) Find the change in h over each half-second interval.

(b) Based on these changes, is the graph of h concave up or concave down on $[0, 2]$? Give a reason.

(c) Between which two recorded times does h change from increasing to decreasing, according to the table?